Validating attribution methods for a frozen pretrained SFCN on pediatric brain MRI

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Abstract

Explanation methods are routinely used to justify clinical claims from 3D brain-MRI classifiers, yet they are rarely validated before being believed. We validate nine attribution methods on a frozen pretrained Simple Fully Convolutional Network (SFCN), a brain-age model, using an age-group proxy task on 96 children from the public Calgary Preschool control cohort. The study makes no claim about prenatal alcohol exposure; it asks which explanations are faithful to the model, how well they agree with one another, and whether region-level claims survive multiple-comparison correction. Under the pre-registered mean-intensity fill, none do. The faithfulness ranking depends on the perturbation fill. Under zero fill, occlusion-based ReX has the lowest deletion AUC (0.14 vs. 0.53–0.78), but under in-distribution mean-intensity fill it scores worst on both deletion and insertion, and under blur fill Guided Backprop leads on insertion. Eight of nine methods passed a randomized-weight sanity check; Integrated Gradients did not (Spearman $r = 0.73$ between trained and random models), indicating input-driven rather than model-driven maps. Methods agreed only weakly (mean pairwise Spearman 0.36), and the two Grad-CAM variants were anti-correlated. The pre-registered region test (200 random same-size regions) found no cortical region under the mean-intensity fill. Under zero fill it flagged 14 regions, all at the null's p-value floor. A post-hoc sensitivity analysis with a 500-draw contiguous null found no region under either fill, so the zero-fill result is not robust to the choice of null. Method-to-masking agreement was weak (Spearman -0.04 to 0.19). Method choice materially changes which explanations look faithful, and occlusion-based faithfulness and randomized-weight sanity checks should be routine. The contribution is a validation template, not a diagnostic classifier.

Keywords: explainable AI, saliency maps, faithfulness, sanity checks, pediatric MRI, brain age, SFCN, prenatal alcohol exposure.

1 Introduction

Deep classifiers for 3D brain MRI are increasingly accompanied by saliency or class-activation maps that purport to show *where* the network looks. When those maps are read anatomically ("the model attends to the corpus callosum"), they migrate quickly into clinical narrative. But attribution methods are known to disagree with each other and, in some cases, to produce the same map whether or not the model has learned anything (Adebayo et al., 2018). Before an explanation is trusted enough to support a claim about a brain region, three questions should be answered: is it *faithful* (do the voxels it highlights actually drive the model?), is it *consistent* (do independent methods agree on which regions matter?), and is it *anatomically valid* under a corrected statistical

test? We use "consistency" in the specific sense of cross-method agreement; we do not test map stability under input perturbation or random seeds, which we flag as a limitation (Section 6).

Das et al. (2024) trained a frozen pretrained SFCN to detect prenatal alcohol exposure (PAE) from pediatric brain MRI and reported explanations as a *preliminary* step: a single Grad-CAM method, with only one named region (white-matter volume) formally tested ($t = -0.73$, $p = 0.47$). The PAE scans are not public. This work does not attempt to reproduce their clinical result. Instead, it takes the identical frozen SFCN backbone and asks the three validation questions above on a task the public data can actually support: an age-group proxy, which is appropriate because the frozen network is a brain-age model and age is the signal its features encode.

Our contributions are: (i) a pre-specified, multi-method validation template for 3D-MRI attribution combining faithfulness, randomized-weight sanity checks, resolution, cross-method agreement, and a corrected anatomical region test; (ii) an empirical verdict over nine methods on a real neuroimaging backbone; and (iii) a public, reproducible pipeline. The headline finding is that *method choice is not a detail*: under zero fill faithfulness spans a factor of five across methods, and the ranking changes with the fill, one widely used method (Integrated Gradients) fails the sanity check, and methods agree only weakly on which regions matter.

The paper follows the three questions as a spine. We answer *faithful?* with deletion and insertion AUC (Section 4.2), *model-driven rather than input-driven?* with a randomized-weight sanity check (Section 4.3), *consistent?* with cross-method agreement (Section 4.5), and *anatomically valid?* with a multiple-comparison-corrected region-masking test (Section 4.6); resolution (Section 4.4) measures whether a map is fine enough to name a structure at all.

2 Related work

2.1 Attribution methods and why they must be validated

Attribution methods assign each input voxel a score for how much it drove a prediction, but they compute that score in very different ways, and the differences matter for what a map can be trusted to show. *Gradient* methods (Saliency (Simonyan et al., 2014), Integrated Gradients (Sundararajan et al., 2017), SmoothGrad (Smilkov et al., 2017), VarGrad (Adebayo et al., 2018)) read the sensitivity of the output to each voxel and so produce voxel-resolution but often noisy maps. *Class-activation* methods (Grad-CAM (Selvaraju et al., 2017)) weight the network's last convolutional feature maps, yielding smooth, coarse localizations at the resolution of that feature grid. *Relevance-propagation* methods (LRP (Bach et al., 2015)) redistribute the output backward through the network under conservation rules. *Perturbation/occlusion* methods (RISE (Petsiuk et al., 2018); here, ReX (Chockler et al., 2024)) instead mask parts of the input and watch the prediction move, so they test the model directly rather than reading its internals. Because these mechanisms differ, the same scan can yield maps that disagree on which region matters, which is exactly what makes a validation step necessary rather than optional.

A body of work shows that a visually convincing map need not reflect anything the model learned. Adebayo et al. (2018) found that several popular methods are largely invariant to randomizing the model's weights, so their output cannot be explaining learned structure; our randomized-weight sanity check follows that logic. Kindermans et al. (2019) showed saliency can change under input shifts that leave the model unchanged, and Ghorbani et al. (2019) that interpretations are fragile to imperceptible perturbations. To move past appearance, two functional criteria are now

standard: occlusion-based faithfulness, measured as the prediction drop when the highlighted voxels are removed (deletion/insertion (Petsiuk et al., 2018)) or as retraining-based importance (ROAR (Hooker et al., 2019)). Critically, this validation literature is almost entirely 2D natural-image classification. The handful of medical-imaging studies that test attribution robustness (Eitel and Ritter, 2019; Arun et al., 2021) report that saliency maps are often unreliable for localization, yet 3D neuroimaging attributions are still routinely read anatomically without any such check. That gap, a full multi-criterion validation of attribution on a real 3D brain-MRI backbone, is what this paper addresses.

2.2 Relation to Das et al. (2024)

This is independent research, carried out by an author not affiliated with the original study, that builds on and extends the explainability step that Das et al. (2024) explicitly flagged as preliminary. The original contribution, the frozen-SFCN transfer-learning approach to prenatal alcohol exposure and its clinical framing, is theirs; our contribution is the validation template applied to the same backbone on public data. It does not re-examine their clinical conclusion, which we cannot access, and every use we make of their design is cited as such. We outline five next steps for extending the original analysis (a finer explanation grid, multi-region testing, site and recruitment controls, age control, and one scan per child), each addressed in Section 5.3 to the extent the public data allow. The complete preregistered protocol (OSF `jycnr`) accompanies this work in the project repository.

3 Methods

3.1 Data and cohort

We use the public Calgary Preschool control cohort (Reynolds et al., 2020) (OSF fork `5kz2p`, CC-BY 4.0), restricted to the 96 children who carry a demographic label and a usable T1 scan, one scan per child. The public release is longitudinal and lists multiple visits per child; its T1 sheet lists 279 scans for these 96 children. We take one scan per child, with child-grouped cross-validation. Cohort demographics are in Table 1. The proxy task is age dichotomized at the cohort median (3.64 y): 48 older (label 1), 48 younger (label 0), a perfectly balanced binary problem (Fig. 9a).

Table 1: Cohort characteristics ($n = 96$ children, one scan each).

Characteristic	Value
N children	96
Age, years, mean (SD)	3.84 (0.83)
Age, years, range	2.58 to 6.90
Male, n (%)	50 (52.1)
Female, n (%)	46 (47.9)

3.2 Preprocessing

Each volume underwent ANTs (Avants et al., 2011) N4 bias correction (Tustison et al., 2010), antspynet deep-learning brain extraction (Tustison et al., 2021), affine registration to 1 mm MNI152,

and cropping to $160 \times 192 \times 160$, the input space in which the SFCN was pretrained. All 96 volumes passed visual QC (Fig. 9b). This is a deliberate design decision and differs from Das et al. (2024), who registered to a pediatric atlas and used FSL-BET + SynthStrip. Because our backbone is frozen and never fine-tuned, its features are only valid on inputs drawn from the adult UK Biobank/MNI152 distribution it was trained on; registering the children into a pediatric template would move the inputs off that distribution and invalidate the very features we are examining. Keeping the model's native space is therefore the methodologically consistent choice for a frozen backbone. Its one cost is that an adult atlas is then applied to preschool brains through affine-only registration, so region-level anatomical labels are approximate; we bound that cost in Section 6.

3.3 Model

We use the original SFCN (Peng et al., 2021), a lightweight fully convolutional brain-age network, with its six convolutional blocks frozen. The frozen backbone uses the UK Biobank brain-age weights released by Peng et al. (2021) in their official repository (Peng, 2021) (checkpoint `brain_age/run_20190719_00_epoch_best_mae.p`); we did not retrain the backbone. A 512-unit head with a 2-logit softmax replaces the paper's PAE head and is trained for the age-group proxy. All attributions are computed with respect to this frozen-backbone, trained-head model, using the head trained on the full 96-child cohort; the cross-validation in Section 4.1 is a separate accuracy estimate and does not feed the attribution model.

3.4 Explanation methods

We compare nine methods (Table 2), spanning gradient, class-activation, relevance-propagation, and occlusion families. M8 (Guided Backprop (Springenberg et al., 2015)) is included as a negative control; M1 (final-block Grad-CAM) is the method used by Das et al. (2024). Captum (Kokhlikyan et al., 2020) provides M1–M6 and M8; zennit (Anders et al., 2021) provides M7; M9 is ReX (Chockler et al., 2024), a 3D occlusion-responsibility search on the $5 \times 6 \times 5$ feature grid. All attributions are computed toward each scan's true proxy-label class, on all 96 children. Integrated Gradients uses a zero-image baseline and 32 steps; SmoothGrad and VarGrad use 20 noised samples ($\sigma = 0.1$); Grad-CAM targets the final (M1) and an earlier (M2) convolutional block.

Table 2: The nine explanation methods.

ID	Method	ID	Method
M1	Grad-CAM, final block (paper's)	M6	VarGrad
M2	Grad-CAM, earlier block	M7	LRP (EpsilonPlusFlat)
M3	Saliency	M8	Guided Backprop (control)
M4	Integrated Gradients (32 steps)	M9	ReX (3D occlusion)
M5	SmoothGrad (20 samples)		

3.5 Metrics and statistics

Each metric below answers one of the three framing questions; two supporting metrics (resolution, and the classifier/probe checks) tell us whether the map and the proxy are even interpretable.

Table 3 is the map from metric to question to pass criterion; the paragraphs that follow give the detail.

Table 3: What each metric measures, the question it answers, and what counts as a pass.

Metric	Question answered	Pass / reading
Faithfulness	Do the highlighted voxels drive the output?	Low deletion AUC, high insertion AUC
Sanity	Is the map model-driven, not input-driven?	$\|r\| < 0.3$ between trained and random model
Agreement	Do independent methods agree on regions?	High pairwise region-level r
Region test	Which regions is the *model* sensitive to, under correction?	Survives BH-FDR $q = 0.05$ vs. the registered random-region null (200 draws)
Resolution	Is the map fine enough to name a structure?	Supporting: top-5% mass, blob count
Classifier / probe	Is the proxy signal real (not noise)?	Supporting: above-chance CV, permutation p

Faithfulness (question 1). Deletion and insertion AUC (Petsiuk et al., 2018) (lower deletion AUC is better). This is a partial-deletion curve, not a full 0–100% sweep: we remove the top-ranked voxels at four fractions (5, 10, 20, 30%) with a zero fill (the registered check fill; the mean-intensity and blur fills were added later, Section 4.2) and integrate the predicted-class probability over that range. We cap at 30% because the discriminative signal is concentrated in the top voxels and deleting a third of the brain already drives most methods to chance; the AUC should be read as "area over 5–30% deletion," not the conventional full-curve value. If a map is faithful, deleting what it ranks highest should collapse the prediction fastest.

Sanity (question 2). Spearman correlation of a method's map between the trained model and a fully reinitialized model (Adebayo et al., 2018); low correlation is good. The reinitialization is complete, not head-only: we deep-copy the model and call `reset_parameters()` on every module that defines it, so *all six convolutional backbone blocks and the classifier head* are reset from a fixed seed. This matters for interpreting the result: a method that stays correlated here is unchanged even when the entire network, including the frozen feature extractor, is random, so its map cannot be reflecting learned structure. We use the standard complete-randomization variant; the layer-by-layer cascading variant of Adebayo et al. (2018) was not run (Section 6). Maps are compared as the per-voxel absolute attribution, with a single Spearman correlation taken over all voxels of the two maps; the random model is a single reinitialization from a fixed seed, so the check uses one random draw rather than an average over seeds. A method is scored PASS when $|r| < 0.3$; this band is a pre-specified heuristic, not a published constant, and is drawn as the dotted line in Fig. 3.

Agreement (question 3, part A). Pairwise Spearman correlation of region-level scores: if the methods are measuring the same underlying signal, their regional rankings should line up.

Region test (question 3, part B). In plain terms, this test asks which brain regions the *model* actually depends on, and it does so *without reference to any saliency map*: for each Harvard–Oxford cortical region (Desikan et al., 2006; Makris et al., 2006) we blank the region's voxels and record the drop in predicted-class probability. A large drop means the model relied on that region. This is why it is a masking test on model output, not an attribution test; it validates the model's regional dependence directly, so it can confirm or refute what the maps claim. Significance is tested against

the pre-registered null of 200 random same-size regions, with Benjamini–Hochberg correction at $q = 0.05$ (Benjamini and Hochberg, 1995) applied across all 96 region-by-fill tests (48 regions under each fill). The primary fill is mean-intensity, as registered; zero fill was registered as a check. The smallest attainable p under this null is $1/201 \approx 0.005$. In addition to the registered 200 draws, we ran a post-hoc sensitivity analysis with 500 same-size regions drawn as translations of the real mask inside gray matter (a contiguous null, smallest attainable p of $1/501 \approx 0.002$). This analysis is exploratory and is listed in Section 5.2. We report both analyses and treat neither as confirmatory for the zero-fill result.

Resolution (supporting). Fraction of attribution mass in the top 5% of voxels, and blob count, how localized a map is, hence whether it is fine enough to name a structure. Blob count is the number of 6-connected components among voxels whose absolute attribution is at or above the map's 95th percentile. It measures fragmentation, not blob size.

Uncertainty and significance. 95% bootstrap CIs (10,000 resamples, resampled by child) on every metric. The classifier check is a repeated (10×) stratified 5-fold CV grouped by child, plus a 1,000-permutation test against chance, establishing that the proxy signal is real before any map is interpreted. All tests were logged with raw and corrected p-values; tests run after seeing the data are flagged exploratory in the pre-specified plan.

4 Results

4.1 Proxy classifier (a sanity check, not a result)

The head need only learn the proxy well enough to make attributions meaningful; its accuracy is not a scientific claim. Cross-validated accuracy was 0.643 ± 0.120 over 50 fits (sensitivity 0.632, specificity 0.668), and a linear probe on the frozen features reached 0.635 (permutation $p = 0.01$, 1,000 permutations; this test comes from the original run, since the revised masking runs set $N_{\text{perm}} = 0$ and skipped it). The proxy is decodable above chance but only modestly (Fig. 1).

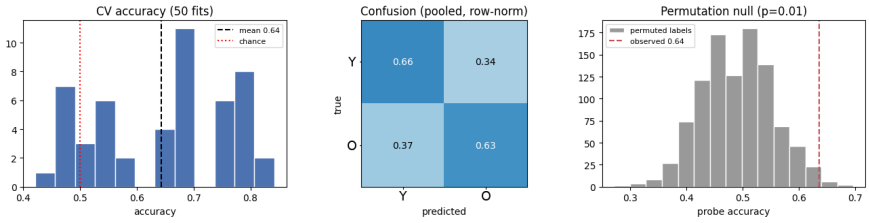


Figure 1: Proxy-task classifier performance (age group from frozen SFCN features).

4.2 Faithfulness

Addressing question 1 (do the highlighted voxels drive the output?), the answer depends on the perturbation fill (Table 4, Appendix C). Under zero fill, ReX (M9) has the lowest deletion AUC (0.141 [0.119, 0.165], against 0.53–0.78 for the rest), with a near-tie with Guided Backprop on insertion. This advantage is partly by construction, since ReX searches for regions whose removal changes the prediction and deletion scores exactly that. Zero blanking is also out of distribution, so the zero-fill ranking may reflect the model's response to blanking rather than faithfulness. Under the in-distribution mean-intensity fill, ReX has the highest deletion AUC (0.877) and the lowest

insertion AUC (0.295). Both are worse than random order here: deletion 0.877 against 0.680 (lower is better), insertion 0.295 against 0.500 (higher is better). The paired ReX-minus-Guided-Backprop insertion difference is −0.195 [−0.265, −0.126]. Under the blur fill, Guided Backprop has the lowest deletion AUC (0.753) and the highest insertion AUC (0.882), and the paired insertion difference is −0.370 [−0.429, −0.311]. Early-block Grad-CAM (M2) has the highest deletion AUC under zero and blur fills. We therefore make no claim that any method is the most faithful. The three fills agree only that deletion and insertion rankings are not stable across fills. The random-order baseline also depends on the fill (deletion 0.592, 0.680 and 0.948 for zero, mean and blur), so deletion AUC is not a fill-independent reference scale. Fig. 2 shows the zero-fill comparison.

Table 4: Faithfulness under three fills. Deletion AUC (lower is better) and insertion AUC (higher is better), all 96 children, same frozen model. Zero fill is the earlier registered-check fill; mean and blur fills were added after the first run (plan section 10). Zero-fill values match the earlier run to within 0.001. The random-order baseline is measured per fill.

	Deletion AUC			Insertion AUC		
Method	Zero	Mean	Blur	Zero	Mean	Blur
M1 Grad-CAM (final)	0.644	0.737	0.860	0.501	0.541	0.614
M2 Grad-CAM (early)	0.777	0.788	0.952	0.488	0.462	0.562
M3 Saliency	0.567	0.708	0.800	0.490	0.458	0.690
M4 Integrated Gradients	0.528	0.604	0.861	0.513	0.500	0.605
M5 SmoothGrad	0.556	0.665	0.794	0.538	0.488	0.677
M6 VarGrad	0.580	0.710	0.821	0.501	0.497	0.638
M7 LRP	0.674	0.789	0.861	0.598	0.527	0.602
M8 Guided Backprop	0.556	0.818	0.753	0.742	0.490	0.882
M9 ReX	0.141	0.877	0.845	0.755	0.295	0.512
Random order (baseline)	0.592	0.680	0.948	0.429	0.500	0.573

Table 5: Faithfulness. Deletion AUC (lower is better) and insertion AUC (higher is better), each with 95% bootstrap CI (10,000 resamples of the 96 children; the child, not the scan, is the resampling unit). Insertion CIs and the random-order baseline come from the academic-account re-run of the faithfulness stage; its insertion values differ from the first run by at most 0.001.

Method	Deletion AUC	95% CI	Insertion AUC	95% CI
M9 ReX	**0.141**	[0.119, 0.165]	0.755	[0.713, 0.793]
M4 Integrated Gradients	0.528	[0.475, 0.584]	0.513	[0.479, 0.548]
M8 Guided Backprop	0.556	[0.489, 0.624]	0.742	[0.701, 0.780]
M5 SmoothGrad	0.556	[0.502, 0.609]	0.538	[0.486, 0.589]
M3 Saliency	0.568	[0.518, 0.619]	0.490	[0.438, 0.543]
M6 VarGrad	0.580	[0.531, 0.628]	0.501	[0.440, 0.560]
M1 Grad-CAM (final)	0.644	[0.601, 0.687]	0.501	[0.471, 0.532]
M7 LRP	0.674	[0.627, 0.720]	0.598	[0.529, 0.665]
M2 Grad-CAM (early)	**0.777**	[0.730, 0.822]	0.488	[0.477, 0.499]
Random order (baseline)	0.592 (SD 0.189)	–	0.429 (SD 0.311)	–

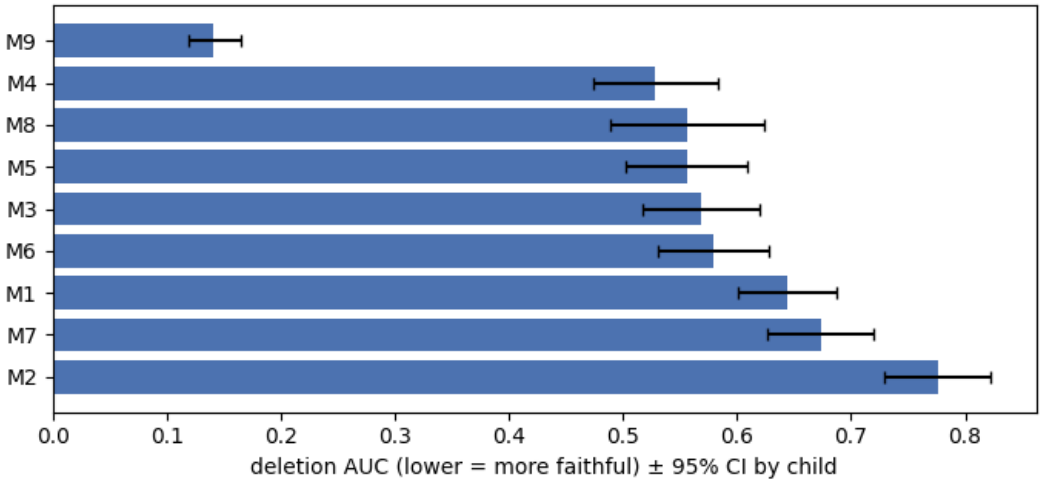


Figure 2: Deletion AUC under zero fill across the nine methods (labels M1–M9 as in Table 2). Lower is more faithful; bars show 95% CIs by child.

4.3 Randomized-weight sanity check

Addressing question 2 (is the map model-driven rather than input-driven?), eight methods pass (low trained-vs-random correlation). Integrated Gradients does not ($r = 0.73$): its map is almost unchanged when every layer, backbone and head, is reinitialized, so on this network it reflects image structure rather than learned content (Table 6, Fig. 3). Because the randomization is complete rather than head-only, this correlation cannot be an artifact of an unchanged frozen backbone: the backbone is random too. Part of this is mechanical and worth stating: with the zero baseline we use, Integrated Gradients reduces to the input-times-gradient form, so a dependence on input intensity is built into the method and survives weight randomization. The practical caution is nonetheless real, because IG is widely used in neuroimaging (Eitel and Ritter, 2019) and its maps look convincing.

Guided Backprop (M8), our negative control, passes here ($r = 0.168$), which is at first surprising: Adebayo et al. (2018) report it as largely invariant to model randomization and therefore a method that typically *fails* this check. The reconciliation cuts two ways, and the second way is a caveat on our own test. The verdict depends on architecture and implementation: Guided Backprop's insensitivity was shown on 2D ImageNet CNNs with ReLU backprop overrides, and on a frozen six-block 3D SFCN with a small trained head the trained-vs-random map is less correlated. But a control passing where the literature expects it to fail also means our single-draw, complete-randomization check is less sensitive here than the cascading variant would be, so the PASS column should be read as a coarse filter, not a certificate. Passing it is in any case necessary, not sufficient: M8 is still a poor explanation on the metric that matters, with a near-chance deletion AUC (0.556).

Table 6: Randomized-weight sanity check. Spearman correlation, trained vs. random model (lower is better).

Method	Spearman (trained vs. random)	Verdict
M2 Grad-CAM (early)	−0.242	PASS
M9 ReX	0.019	PASS
M1 Grad-CAM (final)	0.121	PASS
M3 Saliency	0.141	PASS
M6 VarGrad	0.153	PASS
M8 Guided Backprop	0.168	PASS
M5 SmoothGrad	0.172	PASS
M7 LRP	0.255	PASS
M4 Integrated Gradients	**0.728**	**FAIL (input-driven)**

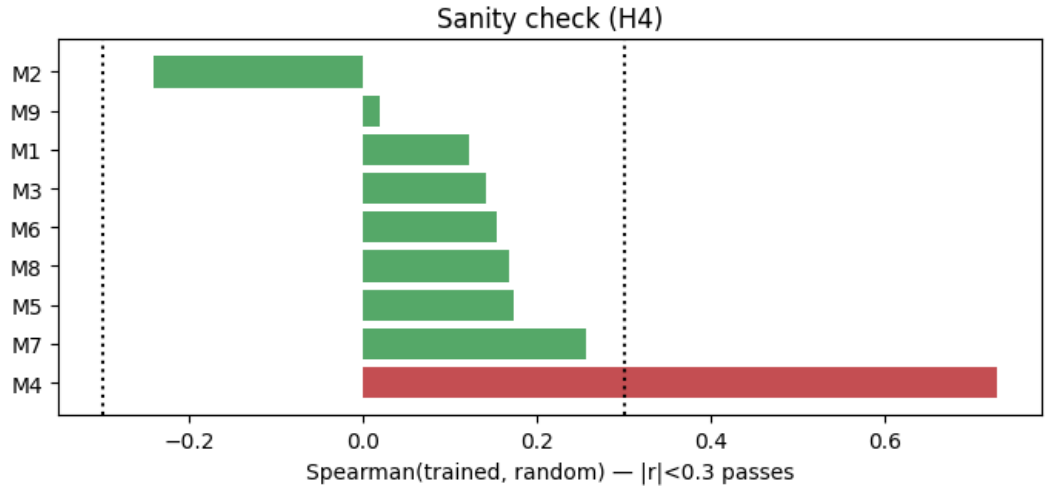


Figure 3: Sanity check: similarity between trained-model and random-model attributions.

4.4 Resolution

The two methods that operate on the $5 \times 6 \times 5$ feature grid, final-block Grad-CAM (M1) and ReX (M9), produce only about three to four large blobs each; in Fig. 4 their blob counts sit near zero against a scale dominated by the voxel-level methods, which fragment into thousands of small blobs (SmoothGrad ≈19,000, Saliency ≈44,000). The near-zero bars for M1 and M9 are therefore expected, not a plotting error: coarse grid-level maps cannot have many blobs. This bears directly on the first next step in Section 5.3: the explanation grid is coarser than many named structures.

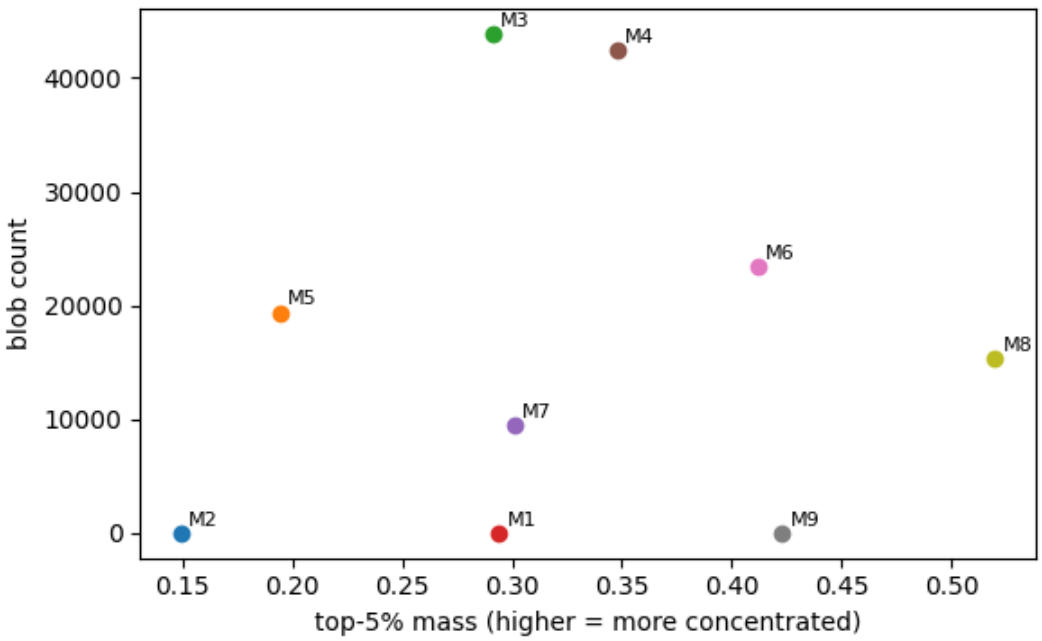


Figure 4: Resolution: top-5% attribution mass and blob count per method. Grid-level methods (M1 Grad-CAM final, M9 ReX) show ≈3–4 blobs, appearing near zero on the voxel-level scale; this is inherent to the $5 \times 6 \times 5$ grid, not an artifact.

4.5 Cross-method agreement

Mean pairwise region-level Spearman correlation over the 36 method pairs is only 0.36 (Fig. 5). The gradient family (M3/M5/M6) agrees strongly among itself, with pairwise r reaching 0.96 at the top of its range (reported as the maximum, not a typical value), but the two Grad-CAM variants are anti-correlated ($r = -0.61$). ReX correlates most with final-block Grad-CAM ($r = 0.56$) and is anti-correlated with early-block Grad-CAM ($r = -0.54$). Which regions one reports depends heavily on the method chosen.

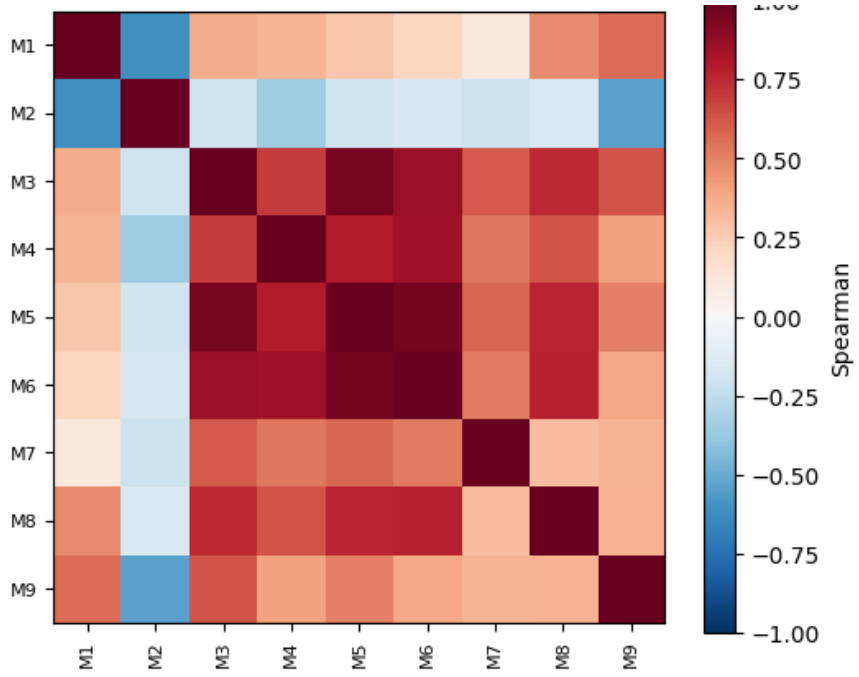


Figure 5: Pairwise cross-method agreement (region-level Spearman correlation).

4.6 Anatomical region test

Registered analysis. Under the mean-intensity fill, no region survives correction at $q = 0.05$ (smallest raw $p = 0.945$). Under the zero-fill check, fourteen regions survive (Table 7). All fourteen have raw $p = 0.005$, the smallest attainable under the 200-draw null, so the correction cannot rank them; their pooled FDR values are all 0.034. One further region (Supramarginal Gyrus, posterior division, raw $p = 0.010$) has FDR 0.064 and does not survive.

Sensitivity analysis. Under the 500-draw contiguous null, no region survives under either fill (Table 8). Under zero fill the smallest FDR p is 0.192 (Lingual Gyrus, raw $p = 0.002$). Under mean fill the smallest raw p is 0.19. The same masking effects are used in both analyses, so the method-to-masking correlations below do not depend on the null.

Interpretation. The zero-fill result depends on the null model: it is significant under the registered null and not under the sensitivity null. Zero blanking is also out of distribution, so part of any zero-fill effect may reflect how much signal is removed rather than the region's specific importance. The mean-fill effects are near zero and not significant under either null. We therefore make no robust anatomical claim.

Method-to-masking agreement. For each method we correlated its mean absolute attribution per region (48 regions) with the zero-fill masking effect (Spearman; permutation p from 10,000 shuffles of the masking effect). Correlations range from -0.04 (M2) to $+0.19$ (M1), with p from 0.20 to 0.96 (`results_revised_null/method_vs_masking_correlation_merged.csv`). The 48 regions are spatially autocorrelated, so these p-values are optimistic. No method's region ranking tracks the model's masking dependence, so we claim no anatomical validity for any method.

Table 7: Regions surviving BH-FDR $q = 0.05$ under the registered null (200 random regions), zero fill, BH pooled over all 96 region-by-fill tests. This is an exploratory check: zero fill was registered as a check, not the primary fill. Effect is the probability drop on masking. All fourteen sit at the null's p floor, so the correction cannot rank them.

Region	Effect	n voxels	Raw p	FDR p
Lingual Gyrus	0.257	26,880	0.005	0.034
Temporal Fusiform Cortex, posterior division	0.111	12,680	0.005	0.034
Insular Cortex	0.081	21,449	0.005	0.034
Parahippocampal Gyrus, posterior division	0.069	5,698	0.005	0.034
Inferior Temporal Gyrus, temporooccipital part	0.063	11,973	0.005	0.034
Cingulate Gyrus, anterior division	0.059	20,844	0.005	0.034
Parahippocampal Gyrus, anterior division	0.057	9,956	0.005	0.034
Cingulate Gyrus, posterior division	0.057	19,228	0.005	0.034
Occipital Fusiform Gyrus	0.053	14,538	0.005	0.034
Central Opercular Cortex	0.044	14,953	0.005	0.034
Parietal Operculum Cortex	0.043	8,739	0.005	0.034
Subcallosal Cortex	0.039	9,021	0.005	0.034
Frontal Operculum Cortex	0.035	5,313	0.005	0.034
Planum Temporale	0.025	8,032	0.005	0.034

Table 8: Sensitivity analysis: region masking under the 500-draw contiguous null, merged over both halves; BH-FDR pooled over all 96 region-by-fill tests. Top five zero-fill and top three mean-fill regions by raw p. None survives $q = 0.05$. Effect is the probability drop on masking; n is region voxel count.

Fill	Region	Effect	n voxels	Raw p	FDR p
zero	Lingual Gyrus	0.257	26,880	0.002	0.192
zero	Parietal Operculum Cortex	0.043	8,739	0.024	0.990
zero	Cingulate Gyrus, anterior div.	0.059	20,844	0.058	0.990
zero	Inferior Temporal Gyrus, temporooccipital	0.063	11,973	0.076	0.990
zero	Planum Temporale	0.025	8,032	0.086	0.990
mean	Precentral Gyrus	0.015	70,031	0.186	0.990
mean	Superior Frontal Gyrus	0.011	45,246	0.206	0.996
mean	Middle Temporal Gyrus, posterior div.	0.007	21,879	0.218	0.996

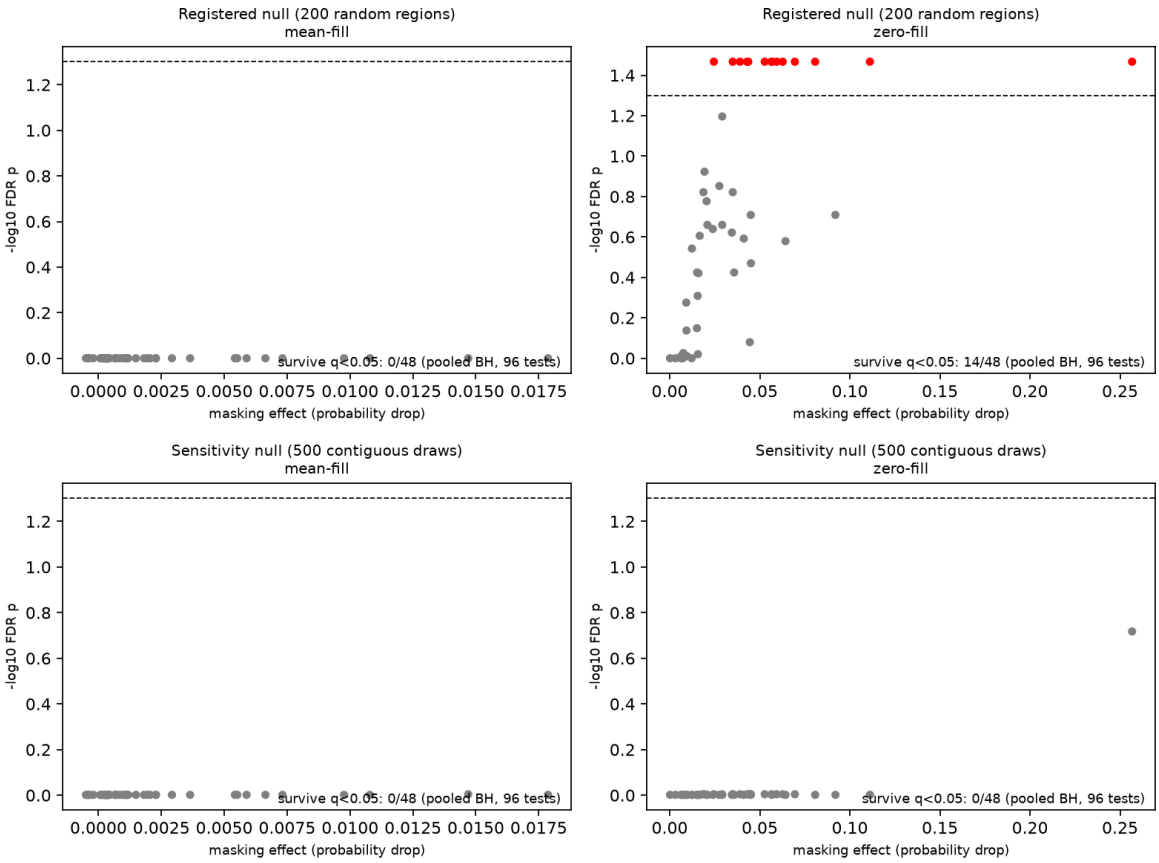


Figure 6: Region masking under the registered null (top, 200 random regions) and the sensitivity null (bottom, 500 contiguous draws), mean fill (left) and zero fill (right). Masking effect vs. $-\log_{10}$ FDR p (BH over 48 regions per fill; pooled over 96 tests). Dashed line marks $q = 0.05$; red points survive the pooled correction.

4.7 Layer probes and qualitative example

Linear-probe accuracy was 0.624 from block 2 and 0.635 from block 6 (Fig. 7): depth adds little for this task, and the proxy is almost as decodable from an early block as from the final one. A representative scan across all nine methods (Fig. 8) illustrates the resolution and agreement differences quantified above.

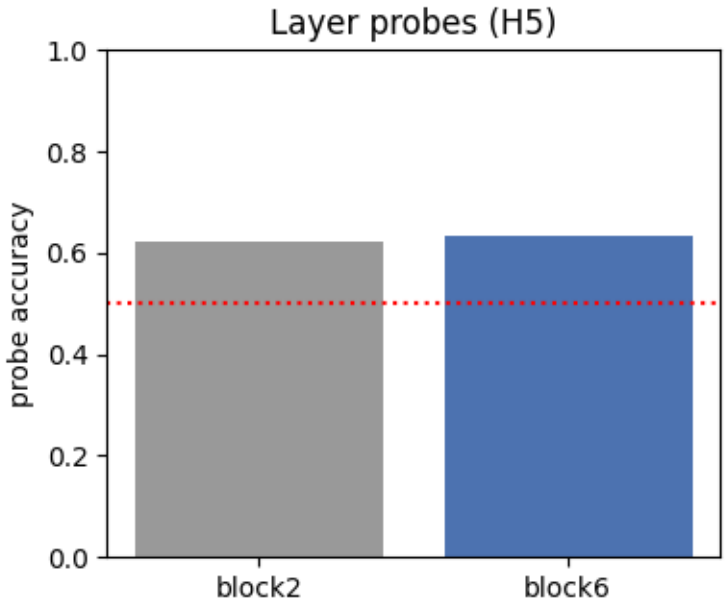


Figure 7: Linear-probe accuracy by feature block (block 2 vs. block 6).

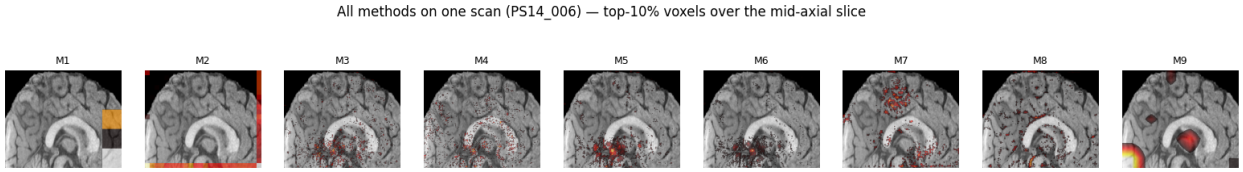


Figure 8: Qualitative attribution maps for one representative scan across all nine methods, on matched slices. The resolution and agreement differences quantified in Figs. 4 and 5 are visible here.

5 Discussion

5.1 Pre-specified hypotheses

Results are reported against the six hypotheses exactly as registered (OSF `jycnr`).

H1 (Grad-CAM produces smoother, coarser maps than Integrated Gradients and SmoothGrad): *supported on blob count, mixed on mass.* Final-block Grad-CAM resolves to ≈3 connected blobs and early-block Grad-CAM to ≈13, versus ≈19,000 for SmoothGrad and ≈42,000 for Integrated Gradients (Fig. 4); this blob-count gap is the unambiguous evidence for "coarser." Top-5% attribution mass does not discriminate cleanly: early-block Grad-CAM (M2) carries the lowest mass, but final-block Grad-CAM (M1, ≈0.30) is *more* concentrated than SmoothGrad (≈0.19), so we rest the claim on blob count rather than mass.

H2 (large anatomical regions are more likely to survive correction than small named structures such as the pons and corpus callosum): *not testable*; under the registered mean-intensity test no region survives BH-FDR ($q = 0.05$), and the 14 zero-fill survivors of the registered null do not survive the sensitivity null (Section 4.6), so the large-versus-small contrast cannot be tested with a confirmatory result.

H3 (Integrated Gradients and SmoothGrad correlate more with each other than either does with final-block Grad-CAM): *supported descriptively*; mean pairwise region-level Spearman is 0.36, and the gradient family (including IG and SmoothGrad) agrees internally (up to $r = 0.96$) while Grad-CAM sits apart. This is reported descriptively; no formal test of the difference between dependent correlations was pre-specified or run.

H4 (for at least one method, randomized-weight maps differ from trained-weight maps): *supported*; most methods change substantially under weight randomization; Integrated Gradients is the lone weak case, retaining the highest trained-versus-random similarity ($r = 0.73$).

H5 (a block-6 linear probe beats a block-2 probe): *not supported*; 0.635 vs. 0.624 is a 0.011 gap, far inside the cross-validation standard deviation (± 0.12), so depth confers no measurable advantage for this proxy and the two probes should be read as tied.

H6 (signal gate: age-group accuracy beats chance under a child-level label-permutation test; near chance would mean the explanations are not interpretable): *supported*; the cross-validated probe predicts the older-versus-younger median split with accuracy 0.635, permutation $p = 0.01$ (1,000 child-level permutations), and the repeated-CV head reaches 0.643 ± 0.120. Age is the proxy target here, not a confound claim; the age-confound limitation is listed separately. 0.635 is the probe accuracy the permutation test refers to, not the CV-head accuracy.

5.2 Deviations and exploratory analyses

The following are not part of the registered hypotheses and are reported as exploratory: (i) the sensitivity null (500 contiguous draws), added after the first run, and the earlier per-fill BH correction, which was replaced by BH pooled over all 96 region-by-fill tests to match the registered scope (the masking test has no method dimension); together with the observation that the zero-fill result depends on the null model; (ii) the random-order baseline and the method-to-masking correlation analysis, both added after the first run; (iii) whether age is independently decodable from the frozen features as a possible confound (Ridge $R^2 = 0.11$); and (iv) fine-tuning the SFCN, which was pre-specified as out of scope and not run. Choices made after inspecting results are flagged as exploratory here and in the project decision log.

5.3 Extending the original analysis

1. **Finer explanation grid.** Next step: a finer grid would let methods name smaller structures. On this backbone, Grad-CAM resolves to a few blobs on the $5 \times 6 \times 5$ grid, coarser than named structures, so this is the first extension to pursue.

2. **Multi-region testing.** Done here: a multi-region, FDR-controlled test was run where a single region was tested before. Under the registered null, 14 regions pass under zero fill (exploratory) and none under mean fill. Under the sensitivity null, none passes. This is a different test from the original white-matter-volume t-test, so we draw no comparison of effect direction or magnitude with Das et al. (2024).

3. **Site and recruitment controls.** Next step: model site and recruitment effects where scan-level site information is available. PAE scans are not public, so this remains untested here.

4. **Age control.** Next step: control for age explicitly. Age is weakly decodable from the frozen features ($R^2 = 0.11$), so it is a plausible covariate to add.

5. **One scan per child.** This study uses one scan per child with child-grouped cross-validation, so each child is one independent unit.

5.4 What this means in practice

Method choice materially changes the reported "important regions." Three practices follow, one per framing question: for faithfulness, report deletion and insertion under more than one perturbation fill, since rankings change with the fill; for model-dependence, run a randomized-weight sanity check and discard methods that fail it (here, Integrated Gradients); and for anatomical validity, never report a single method's regions without a corrected statistical test.

6 Limitations

$n = 96$; a single proxy task; one cortical atlas; and no PAE data. One caution follows from the native-space design justified in the Methods: the adult Harvard–Oxford atlas reaches the preschool brains through affine-only registration, so for children aged 2.6–6.9 y the region identities in Tables 7 and 8 are approximate anatomical labels, and the anatomical-validity question is answered as "not established" at the resolution of the masking test, not for individual gyri. Age-appropriate pediatric templates and nonlinear registration would sharpen the labels (Fonov et al., 2011) but would invalidate the frozen features, so the trade-off is deliberate. The backbone weights are the public UK Biobank brain-age release of Peng et al. (2021) (Peng, 2021); we report the checkpoint used rather than relying on an unverified source. The proxy head is weak (≈64% CV accuracy), and faithfulness and sanity verdicts are properties of the model being explained: a stronger classifier could rank the methods differently, so the rankings here should be read as holding for this modest-accuracy model, not as universal. At $n = 96$ the study is powered for medium-to-large region effects; small effects may be inconclusive (H3 and H4 do not depend on effect size). Three gaps bound the interpretation. First, we measured cross-method *agreement* but not true map *stability*: we did not quantify how a single method's map varies under small input perturbations or across random seeds, so claims about reproducibility within a method are out of scope. Second, our sanity check uses complete model randomization; we did not perform the layer-by-layer cascading randomization of Adebayo et al. (2018), which would localize where in the backbone a method's dependence breaks down. Third, the random-order baseline is measured (deletion AUC 0.592, insertion AUC 0.429), not assumed to be 0.5; it orders voxels at random, which is a weaker null than a sparsity-matched random map. The registered region test has a p floor of about 0.005 (200 draws), so its 14 zero-fill survivors cannot be ranked. The post-hoc sensitivity analysis (500 draws, floor about 0.002) finds none (Section 4.6). Method-to-masking correlations use the same 48 regions, so they inherit spatial autocorrelation. This work validates an explanation protocol on an age-group proxy in public controls; it does not speak to alcohol exposure, FASD, or the original paper's clinical regions, which require the private PAE data.

7 Conclusion

On a frozen pretrained SFCN, nine explanation methods gave sharply different accounts of what the model uses. Faithfulness rankings depended on the perturbation fill and no method was consistently most faithful; Integrated Gradients failed a basic sanity check; methods agreed only weakly; and

region-level claims were not robust. Under the pre-registered mean-intensity test no region survived correction. The zero-fill check flagged 14 regions under the registered null, but none survived a post-hoc sensitivity null. This rests on one frozen backbone, one head and one cohort, so the protocol is a template to reuse and test, not a settled procedure. The practical message for neuroimaging XAI is that attribution maps must be validated for faithfulness, model-dependence, and anatomical significance, before any of them is read as evidence about the brain.

Acknowledgments

This work builds on Das et al. (2024), whose study motivated the present validation and whose frozen-SFCN transfer-learning design we reuse on public data; we thank those authors for clearly flagging their explainability step as preliminary, which is what this paper extends. We also thank Reynolds et al. (2020) and the Calgary Preschool MRI study for releasing the control cohort under CC-BY 4.0, and Peng et al. (2021) for the public SFCN brain-age weights. The author declares no affiliation with the original study and no competing interests.

Reproducibility

Data: public Calgary Preschool controls (OSF fork `5kz2p`, CC-BY 4.0). Code, the pre-specified plan, all figures, and the full per-region p-value log are in the project repository: `https://github.com/HoussemDs/brain-xai-faithfulness`. The sensitivity masking analysis (merged halves, pooled FDR, method–masking correlations, region figures) is reproduced by `code/merge_masking_halves.py` from the Kaggle outputs, with results in `results_revised_null/`. The full pipeline ran on Kaggle (GPU T4 $\times 2$, $\approx$5.8 h).

References

Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

Christopher J. Anders, David Neumann, Wojciech Samek, Klaus-Robert Müller, and Sebastian Lapuschkin. Software for dataset-wide XAI: from local explanations to global insights with Zennit, CoRelAy, and ViRelAy. *arXiv preprint arXiv:2106.13200*, 2021.

Nishanth Arun, Nathan Gaw, Praveer Singh, Ken Chang, Mehak Aggarwal, Bryan Chen, Katharina Hoebel, Sharut Gupta, Jay Patel, Mishka Gidwani, Julius Adebayo, Matthew D. Li, and Jayashree Kalpathy-Cramer. Assessing the trustworthiness of saliency maps for localizing abnormalities in medical imaging. *Radiology: Artificial Intelligence*, 3(6):e200267, 2021. doi: 10.1148/ryai.2021200267.

Brian B. Avants, Nicholas J. Tustison, Gang Song, Philip A. Cook, Arno Klein, and James C. Gee. A reproducible evaluation of ANTs similarity metric performance in brain image registration. *NeuroImage*, 54(3):2033–2044, 2011. doi: 10.1016/j.neuroimage.2010.09.025.

Sebastian Bach, Alexander Binder, Grégoire Montavon, Frederick Klauschen, Klaus-Robert Müller, and Wojciech Samek. On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation. *PLOS ONE*, 10(7):e0130140, 2015. doi: 10.1371/journal.pone.0130140.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995.

Hana Chockler, David A. Kelly, Daniel Kroening, and Youcheng Sun. Causal explanations for image classifiers. *arXiv preprint arXiv:2411.08875*, 2024.

Anik Das, Kaue Duarte, Catherine Lebel, and Mariana Bento. Deep learning for detecting prenatal alcohol exposure in pediatric brain MRI: a transfer learning approach with explainability insights. *Frontiers in Computational Neuroscience*, 18:1434421, 2024. doi: 10.3389/fncom.2024.1434421.

Rahul S. Desikan, Florent Ségonne, Bruce Fischl, Brian T. Quinn, Bradford C. Dickerson, Deborah Blacker, Randy L. Buckner, Anders M. Dale, R. Paul Maguire, Bradley T. Hyman, Marilyn S. Albert, and Ronald J. Killiany. An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *NeuroImage*, 31(3):968–980, 2006. doi: 10.1016/j.neuroimage.2006.01.021.

Fabian Eitel and Kerstin Ritter. Testing the robustness of attribution methods for convolutional neural networks in MRI-based Alzheimer's disease classification. In *Interpretability of Machine Intelligence in Medical Image Computing (iMIMIC), MICCAI Workshop*, volume 11797 of *Lecture Notes in Computer Science*, pages 3–11. Springer, 2019. doi: 10.1007/978-3-030-33850-3_1.

Vladimir Fonov, Alan C. Evans, Kelly Botteron, C. Robert Almli, Robert C. McKinstry, D. Louis Collins, and Brain Development Cooperative Group. Unbiased average age-appropriate atlases for pediatric studies. *NeuroImage*, 54(1):313–327, 2011. doi: 10.1016/j.neuroimage.2010.07.033.

Amirata Ghorbani, Abubakar Abid, and James Zou. Interpretation of neural networks is fragile. In *AAAI Conference on Artificial Intelligence*, volume 33, pages 3681–3688, 2019. doi: 10.1609/aaai.v33i01.33013681.

Sara Hooker, Dumitru Erhan, Pieter-Jan Kindermans, and Been Kim. A benchmark for interpretability methods in deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

Pieter-Jan Kindermans, Sara Hooker, Julius Adebayo, Maximilian Alber, Kristof T. Schütt, Sven Dähne, Dumitru Erhan, and Been Kim. The (un)reliability of saliency methods. In *Explainable AI: Interpreting, Explaining and Visualizing Deep Learning*, volume 11700 of *Lecture Notes in Computer Science*, pages 267–280. Springer, 2019. doi: 10.1007/978-3-030-28954-6_14.

Narine Kokhlikyan, Vivek Miglani, Miguel Martin, Edward Wang, Bilal Alsallakh, Jonathan Reynolds, Alexander Melnikov, Natalia Kliushkina, Carlos Araya, Siqi Yan, and Orion Reblitz-Richardson. Captum: A unified and generic model interpretability library for PyTorch. *arXiv preprint arXiv:2009.07896*, 2020.

Nikos Makris, Jill M. Goldstein, David Kennedy, Steven M. Hodge, Verne S. Caviness, Stephen V. Faraone, Ming T. Tsuang, and Larry J. Seidman. Decreased volume of left and total anterior insular lobule in schizophrenia. *Schizophrenia Research*, 83(2-3):155–171, 2006.

Han Peng. UKBiobank_deep_pretrain: pretrained SFCN brain-age models. `https://github.com/ha-ha-ha-han/UKBiobank_deep_pretrain`, 2021. Accessed 2026-11-06.

Han Peng, Weikang Gong, Christian F. Beckmann, Andrea Vedaldi, and Stephen M. Smith. Accurate brain age prediction with lightweight deep neural networks. *Medical Image Analysis*, 68: 101871, 2021. doi: 10.1016/j.media.2020.101871.

Vitali Petsiuk, Abir Das, and Kate Saenko. RISE: Randomized input sampling for explanation of black-box models. *British Machine Vision Conference (BMVC)*, 2018.

Jess E. Reynolds, Xiangyu Long, Melody N. Grohs, Deborah Dewey, and Catherine Lebel. Calgary preschool magnetic resonance imaging (MRI) dataset. *Data in Brief*, 29:105224, 2020. doi: 10.1016/j.dib.2020.105224.

Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.

Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. Deep inside convolutional networks: Visualising image classification models and saliency maps. *arXiv preprint arXiv:1312.6034*, 2014.

Daniel Smilkov, Nikhil Thorat, Been Kim, Fernanda Viégas, and Martin Wattenberg. Smoothgrad: removing noise by adding noise. *arXiv preprint arXiv:1706.03825*, 2017.

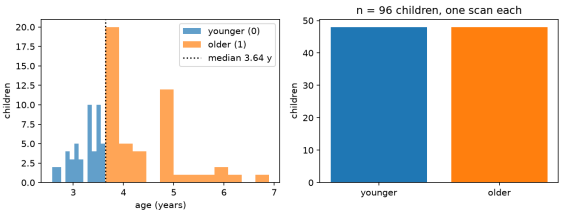
Jost Tobias Springenberg, Alexey Dosovitskiy, Thomas Brox, and Martin Riedmiller. Striving for simplicity: the all convolutional net. In *International Conference on Learning Representations (ICLR) Workshop*, 2015. arXiv:1412.6806.

Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *International Conference on Machine Learning (ICML)*, 2017.

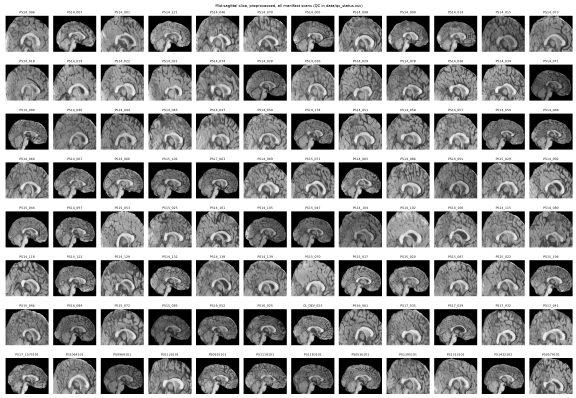
Nicholas J. Tustison, Brian B. Avants, Philip A. Cook, Yuanjie Zheng, Alexander Egan, Paul A. Yushkevich, and James C. Gee. N4ITK: Improved N3 bias correction. *IEEE Transactions on Medical Imaging*, 29(6):1310–1320, 2010. doi: 10.1109/TMI.2010.2046908.

Nicholas J. Tustison, Philip A. Cook, Andrew J. Holbrook, Hans J. Johnson, John Muschelli, Gabriel A. Devenyi, Jeffrey T. Duda, Sandhitsu R. Das, Nicholas C. Cullen, Daniel L. Gillen, Michael A. Yassa, James R. Stone, James C. Gee, and Brian B. Avants. The ANTsX ecosystem for quantitative biological and medical imaging. *Scientific Reports*, 11:9068, 2021. doi: 10.1038/s41598-021-87564-6.

A Supplementary preprocessing figures



(a) Age-group split at the cohort median.



(b) QC contact sheet (all 96 scans).

Figure 9: Cohort and quality control.

B Full region-masking results

Every Harvard–Oxford cortical region (48), both fills, both null models. Regions are ordered by zero-fill raw p under the registered null. Effect is the probability drop on masking (positive = the model depends on the region). BH-FDR is pooled over all 96 region-by-fill tests. The registered null uses 200 random same-size regions (smallest attainable $p \approx 0.005$). The sensitivity null uses 500 contiguous translated regions (smallest $p \approx 0.002$). Machine-readable versions: `kaggle_output_prenatal_xai_96/results/region_masking_all_pvalues.csv` `results_revised_null/region_masking_merged.csv`.

Table 9: Full region-masking results under the registered null (200 random regions).

	Zero fill			Mean fill		
Region	Effect	Raw p	FDR p	Effect	Raw p	FDR p
Insular Cortex	0.081	0.005	0.034	-0.000	1.000	1.000
Inferior Temporal Gyrus, temporooccipital part	0.063	0.005	0.034	+0.002	0.990	1.000
Subcallosal Cortex	0.039	0.005	0.034	+0.001	0.990	1.000
Cingulate Gyrus, anterior division	0.059	0.005	0.034	+0.001	1.000	1.000
Cingulate Gyrus, posterior division	0.057	0.005	0.034	+0.000	1.000	1.000
Parahippocampal Gyrus, anterior division	0.057	0.005	0.034	+0.001	0.995	1.000
Parahippocampal Gyrus, posterior division	0.069	0.005	0.034	+0.001	0.985	1.000
Lingual Gyrus	0.257	0.005	0.034	+0.001	1.000	1.000
Temporal Fusiform Cortex, posterior division	0.111	0.005	0.034	+0.001	1.000	1.000
Occipital Fusiform Gyrus	0.053	0.005	0.034	+0.002	1.000	1.000
Frontal Operculum Cortex	0.035	0.005	0.034	+0.000	0.990	1.000

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	Zero fill			Mean fill		
Region	Effect	Raw p	FDR p	Effect	Raw p	FDR p
Central Opercular Cortex	0.044	0.005	0.034	-0.000	1.000	1.000
Parietal Operculum Cortex	0.043	0.005	0.034	+0.001	0.995	1.000
Planum Temporale	0.025	0.005	0.034	+0.000	1.000	1.000
Supramarginal Gyrus, posterior division	0.029	0.010	0.064	-0.000	1.000	1.000
Frontal Medial Cortex	0.019	0.020	0.119	+0.001	0.980	1.000
Inferior Temporal Gyrus, posterior division	0.028	0.025	0.140	+0.004	0.995	1.000
Superior Temporal Gyrus, posterior division	0.019	0.030	0.151	+0.000	0.995	1.000
Paracingulate Gyrus	0.035	0.030	0.151	+0.000	1.000	1.000
Intracalcarine Cortex	0.020	0.035	0.167	+0.000	1.000	1.000
Middle Frontal Gyrus	0.045	0.045	0.195	+0.007	1.000	1.000
Precentral Gyrus	0.092	0.045	0.195	+0.015	1.000	1.000
Frontal Orbital Cortex	0.029	0.055	0.219	+0.002	1.000	1.000
Temporal Occipital Fusiform Cortex	0.021	0.055	0.219	+0.005	0.945	1.000
Middle Temporal Gyrus, posterior division	0.024	0.060	0.229	+0.007	0.995	1.000
Lateral Occipital Cortex, inferior division	0.034	0.065	0.239	+0.000	1.000	1.000
Inferior Frontal Gyrus, pars triangularis	0.017	0.070	0.248	+0.002	0.980	1.000
Temporal Pole	0.041	0.075	0.256	+0.006	1.000	1.000
Lateral Occipital Cortex, superior division	0.064	0.080	0.264	+0.010	1.000	1.000
Superior Temporal Gyrus, anterior division	0.012	0.090	0.287	+0.000	0.990	1.000
Postcentral Gyrus	0.045	0.109	0.339	+0.006	1.000	1.000
Superior Frontal Gyrus	0.036	0.129	0.376	+0.011	1.000	1.000
Juxtapositional Lobule Cortex (formerly Supplementary Motor Cortex)	0.015	0.129	0.376	+0.000	1.000	1.000
Supramarginal Gyrus, anterior division	0.016	0.134	0.379	+0.000	1.000	1.000
Middle Temporal Gyrus, temporooccipital part	0.016	0.179	0.491	+0.000	1.000	1.000
Heschl's Gyrus (both subdivisions)	0.009	0.199	0.531	+0.000	1.000	1.000
Superior Parietal Lobule	0.015	0.274	0.710	+0.001	1.000	1.000
Middle Temporal Gyrus, anterior division	0.009	0.289	0.729	-0.000	0.980	1.000
Frontal Pole	0.044	0.338	0.833	+0.018	1.000	1.000
Planum Polare	0.008	0.393	0.943	+0.000	0.995	1.000
Occipital Pole	0.016	0.408	0.955	+0.002	1.000	1.000
Inferior Frontal Gyrus, pars opercularis	0.010	0.433	0.977	+0.001	0.990	1.000
Temporal Fusiform Cortex, anterior division	0.006	0.438	0.977	+0.000	0.990	1.000
Precuneous Cortex	0.012	0.507	1.000	+0.003	1.000	1.000
Angular Gyrus	0.007	0.537	1.000	-0.000	1.000	1.000
Cuneal Cortex	0.006	0.562	1.000	+0.000	1.000	1.000
Inferior Temporal Gyrus, anterior division	0.003	0.622	1.000	+0.001	0.945	1.000
Supracalcarine Cortex	0.000	0.945	1.000	+0.000	0.960	1.000

Table 10: Full region-masking results under the sensitivity null (500 contiguous draws).

	Zero fill			Mean fill		
Region	Effect	Raw p	FDR p	Effect	Raw p	FDR p
Insular Cortex	0.081	0.445	0.996	-0.000	0.972	0.996
Inferior Temporal Gyrus, temporooccipital part	0.063	0.076	0.990	+0.002	0.497	0.996
Subcallosal Cortex	0.039	0.152	0.990	+0.001	0.581	0.996
Cingulate Gyrus, anterior division	0.059	0.058	0.990	+0.001	0.830	0.996
Cingulate Gyrus, posterior division	0.057	0.112	0.990	+0.000	0.876	0.996
Parahippocampal Gyrus, anterior division	0.057	0.345	0.996	+0.001	0.786	0.996
Parahippocampal Gyrus, posterior division	0.069	0.182	0.990	+0.001	0.679	0.996
Lingual Gyrus	0.257	0.002	0.192	+0.001	0.912	0.996
Temporal Fusiform Cortex, posterior division	0.111	0.242	0.996	+0.001	0.898	0.996
Occipital Fusiform Gyrus	0.053	0.186	0.990	+0.002	0.571	0.996
Frontal Operculum Cortex	0.035	0.158	0.990	+0.000	0.778	0.996
Central Opercular Cortex	0.044	0.112	0.990	-0.000	0.950	0.996
Parietal Operculum Cortex	0.043	0.024	0.990	+0.001	0.741	0.996
Planum Temporale	0.025	0.086	0.990	+0.000	0.956	0.996
Supramarginal Gyrus, posterior division	0.029	0.178	0.990	-0.000	0.972	0.996
Frontal Medial Cortex	0.019	0.092	0.990	+0.001	0.752	0.996
Inferior Temporal Gyrus, posterior division	0.028	0.357	0.996	+0.004	0.425	0.996
Superior Temporal Gyrus, posterior division	0.019	0.094	0.990	+0.000	0.956	0.996
Paracingulate Gyrus	0.035	0.152	0.990	+0.000	0.952	0.996
Intracalcarine Cortex	0.020	0.485	0.996	+0.000	0.934	0.996
Middle Frontal Gyrus	0.045	0.160	0.990	+0.007	0.248	0.996
Precentral Gyrus	0.092	0.772	0.996	+0.015	0.186	0.990
Frontal Orbital Cortex	0.029	0.469	0.996	+0.002	0.882	0.996
Temporal Occipital Fusiform Cortex	0.021	0.501	0.996	+0.005	0.295	0.996
Middle Temporal Gyrus, posterior division	0.024	0.289	0.996	+0.007	0.218	0.996
Lateral Occipital Cortex, inferior division	0.034	0.373	0.996	+0.000	0.974	0.996
Inferior Frontal Gyrus, pars triangularis	0.017	0.265	0.996	+0.002	0.593	0.996
Temporal Pole	0.041	0.463	0.996	+0.006	0.623	0.996
Lateral Occipital Cortex, superior division	0.064	0.571	0.996	+0.010	0.271	0.996
Superior Temporal Gyrus, anterior division	0.012	0.327	0.996	+0.000	0.894	0.996
Postcentral Gyrus	0.045	0.711	0.996	+0.006	0.411	0.996
Superior Frontal Gyrus	0.036	0.251	0.996	+0.011	0.206	0.996
Juxtapositional Lobule Cortex (formerly Supplementary Motor Cortex)	0.015	0.465	0.996	+0.000	0.894	0.996
Supramarginal Gyrus, anterior division	0.016	0.309	0.996	+0.000	0.960	0.996
Middle Temporal Gyrus, temporooccipital part	0.016	0.561	0.996	+0.000	0.996	0.996
Heschl's Gyrus (both subdivisions)	0.009	0.463	0.996	+0.000	0.888	0.996
Superior Parietal Lobule	0.015	0.758	0.996	+0.001	0.790	0.996

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	Zero fill			Mean fill		
Region	Effect	Raw p	FDR p	Effect	Raw p	FDR p
Middle Temporal Gyrus, anterior division	0.009	0.579	0.996	-0.000	0.886	0.996
Frontal Pole	0.044	0.112	0.990	+0.018	0.890	0.996
Planum Polare	0.008	0.583	0.996	+0.000	0.834	0.996
Occipital Pole	0.016	0.715	0.996	+0.002	0.669	0.996
Inferior Frontal Gyrus, pars opercularis	0.010	0.615	0.996	+0.001	0.743	0.996
Temporal Fusiform Cortex, anterior division	0.006	0.890	0.996	+0.000	0.890	0.996
Precuneous Cortex	0.012	0.745	0.996	+0.003	0.671	0.996
Angular Gyrus	0.007	0.715	0.996	-0.000	0.994	0.996
Cuneal Cortex	0.006	0.810	0.996	+0.000	0.918	0.996
Inferior Temporal Gyrus, anterior division	0.003	0.876	0.996	+0.001	0.735	0.996
Supracalcarine Cortex	0.000	0.990	0.996	+0.000	0.749	0.996

C Faithfulness under three fills, with confidence intervals

95% bootstrap intervals by child (10,000 resamples). Paired insertion difference, ReX minus Guided Backprop, per fill: zero fill +0.013 [-0.046, +0.072]; mean fill -0.195 [-0.265, -0.126]; blur fill -0.370 [-0.429, -0.311]. Machine-readable: `results_revised_null/faithfulness_ci_by_fill.csv`.

Table 11: Deletion and insertion AUC with 95% bootstrap CIs by child, per fill.

Fill	Method	Deletion AUC [95% CI]	Insertion AUC [95% CI]
Zero	M1 Grad-CAM (final)	0.644 [0.601, 0.687]	0.501 [0.471, 0.532]
Zero	M2 Grad-CAM (early)	0.777 [0.729, 0.822]	0.488 [0.477, 0.499]
Zero	M3 Saliency	0.567 [0.517, 0.619]	0.490 [0.438, 0.543]
Zero	M4 Integrated Gradients	0.528 [0.475, 0.584]	0.513 [0.479, 0.548]
Zero	M5 SmoothGrad	0.556 [0.503, 0.609]	0.538 [0.486, 0.589]
Zero	M6 VarGrad	0.580 [0.531, 0.628]	0.501 [0.440, 0.560]
Zero	M7 LRP	0.674 [0.627, 0.720]	0.598 [0.529, 0.665]
Zero	M8 Guided Backprop	0.556 [0.489, 0.624]	0.742 [0.701, 0.780]
Zero	M9 ReX	0.141 [0.119, 0.165]	0.755 [0.713, 0.793]
Zero	Random order	0.592 [0.555, 0.630]	0.429 [0.368, 0.493]
Mean	M1 Grad-CAM (final)	0.737 [0.685, 0.787]	0.541 [0.475, 0.607]
Mean	M2 Grad-CAM (early)	0.788 [0.744, 0.830]	0.462 [0.389, 0.532]
Mean	M3 Saliency	0.708 [0.632, 0.782]	0.458 [0.376, 0.540]
Mean	M4 Integrated Gradients	0.604 [0.521, 0.685]	0.500 [0.404, 0.589]
Mean	M5 SmoothGrad	0.665 [0.596, 0.732]	0.488 [0.399, 0.575]
Mean	M6 VarGrad	0.710 [0.644, 0.773]	0.497 [0.409, 0.585]
Mean	M7 LRP	0.789 [0.725, 0.851]	0.527 [0.446, 0.606]
Mean	M8 Guided Backprop	0.818 [0.769, 0.866]	0.490 [0.397, 0.581]
Mean	M9 ReX	0.877 [0.838, 0.912]	0.295 [0.236, 0.357]
Mean	Random order	0.680 [0.596, 0.763]	0.500 [0.401, 0.589]
Blur	M1 Grad-CAM (final)	0.860 [0.824, 0.893]	0.614 [0.544, 0.681]
Blur	M2 Grad-CAM (early)	0.952 [0.940, 0.963]	0.562 [0.481, 0.644]
Blur	M3 Saliency	0.800 [0.748, 0.849]	0.690 [0.631, 0.747]
Blur	M4 Integrated Gradients	0.861 [0.823, 0.896]	0.605 [0.532, 0.676]
Blur	M5 SmoothGrad	0.794 [0.740, 0.844]	0.677 [0.614, 0.739]
Blur	M6 VarGrad	0.821 [0.775, 0.864]	0.638 [0.569, 0.705]
Blur	M7 LRP	0.861 [0.821, 0.898]	0.602 [0.533, 0.669]
Blur	M8 Guided Backprop	0.753 [0.697, 0.807]	0.882 [0.853, 0.909]
Blur	M9 ReX	0.845 [0.796, 0.891]	0.512 [0.438, 0.587]
Blur	Random order	0.948 [0.935, 0.959]	0.573 [0.489, 0.657]

D Cross-method agreement matrix

Pairwise region-level Spearman correlation between the nine methods (48 regions; each cell is the correlation between two methods' region-score vectors). Values are as reported in the agreement analysis, to two decimals. The mean over the 36 off-diagonal pairs (upper triangle) is 0.36.

Table 12: Pairwise cross-method agreement (Spearman, region level).

	M1	M2	M3	M4	M5	M6	M7	M8	M9
M1	1.00	-0.61	+0.37	+0.34	+0.28	+0.21	+0.10	+0.47	+0.56
M2	-0.61	1.00	-0.21	-0.36	-0.20	-0.17	-0.21	-0.15	-0.54
M3	+0.37	-0.21	1.00	+0.70	+0.95	+0.86	+0.62	+0.74	+0.63
M4	+0.34	-0.36	+0.70	1.00	+0.80	+0.85	+0.53	+0.63	+0.40
M5	+0.28	-0.20	+0.95	+0.80	1.00	+0.96	+0.57	+0.76	+0.50
M6	+0.21	-0.17	+0.86	+0.85	+0.96	1.00	+0.52	+0.77	+0.39
M7	+0.10	-0.21	+0.62	+0.53	+0.57	+0.52	1.00	+0.31	+0.34
M8	+0.47	-0.15	+0.74	+0.63	+0.76	+0.77	+0.31	1.00	+0.35
M9	+0.56	-0.54	+0.63	+0.40	+0.50	+0.39	+0.34	+0.35	1.00